# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for enterprise systems, focusing on robust error handling and asynchronous processing of financial transactions.
- Integrated a **JPA persistence layer** with a PostgreSQL database, managing complex object-relational mappings and ensuring data integrity across distributed services.
- Wrote comprehensive **JUnit and Mockito** test suites for Java applications, achieving over 85% code coverage and identifying critical edge cases before deployment.
- Contributed to the research and development of tools to assist in the **vulnerability discovery process**, automating the identification of common security flaws.
- Collaborated with other teams within **Enterprise Cybersecurity** to improve the overall security of applications and infrastructure, participating in cross-functional security reviews.
- Replicated actual techniques and tools used by malicious attackers to model potential external threats on **web and mobile applications**, identifying key attack vectors.
- Containerized backend services using **Docker** and deployed them to **AWS EC2** instances, configuring basic resource limits and health checks for improved stability.

## SKILLS

**Languages** – Java, Python, SQL, R, TypeScript

**Frameworks/Libraries** – Spring Boot, JPA, JUnit, Mockito, pandas, NumPy, scikit-learn, Flask, FastAPI, React, Angular

**Cloud/DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Kubernetes

**Databases** – PostgreSQL, MySQL, Oracle

**Tools/Platforms** – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau, RStudio

**Methodologies** – Agile, Scrum, CI/CD

## PROJECTS

### Application Security Vulnerability Scanner (HCL Technologies)

- Developed a Python-based tool to automate the scanning of web applications for common security vulnerabilities, such as SQL injection and Cross-Site Scripting (XSS).
- Integrated OWASP ZAP API to systematically probe applications and identify potential security weaknesses, reducing manual testing time by approximately 40%.
- Designed a reporting module that generated detailed vulnerability reports, including remediation steps, for application owners and development teams.

### Secure API Gateway Implementation (MSc Project)

- Built a secure API gateway using Java Spring Boot and Spring Security to manage access control and enforce security policies for downstream microservices.
- Implemented authentication mechanisms including JWT token validation and authorization checks, ensuring only authenticated and authorized users could access protected resources.
- Configured rate limiting and request validation to protect against denial-of-service attacks and malformed input.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)